

Euclid's Elements — Book XI

Proposition 26

Formal Reconstruction — Technical Documentation

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CGJTEAMLAB

To construct a solid angle equal to a given solid angle on a given straight line, and at a given point on it.

1 Purpose and Scope

Proposition XI.26 is the constructive transition from the theory of spatial incidence, perpendicularity, and congruence developed earlier in Book XI to the explicit copying of a complete trihedral angle. The input consists of a given trihedral angle at a point D , together with a prescribed straight line AB and a prescribed point A on that line. The goal is to construct at A a new trihedral angle whose three plane face angles are respectively congruent to the three face angles of the source trihedron.

Classically the construction is short. Euclid projects a point on one edge of the source trihedron perpendicularly to the plane of the other two edges, copies the resulting planar data at the prescribed point, erects a perpendicular to the copied base plane, and recovers the remaining face angles by triangle congruence [?].

The formal reconstruction reveals that the short classical proof suppresses a substantial order problem. If G is the perpendicular projection of the third source point F to the base plane through D, E, C , then the ray DG need not lie in the interior of the base angle EDC . It may coincide with the vertex, lie on either boundary ray, continue beyond a boundary ray, lie in either adjacent exterior sector, lie in the interior sector, or lie in the remaining cyclic sector. A complete theorem must construct the copied point in every one of these positions.

The production proof therefore has two conceptual layers:

1. a proposition-specific planar construction and complete order classification of the projection foot;
2. a shared spatial metric layer using perpendicular-to-plane data, right triangles, SAS, SSS, and the spatial form of Hilbert's angle-addition theorem.

The direct production theorem is

```
Geometry.euclid_proposition_11_26
```

in

```
CGJteamLab/Proposition11_26.lean
```

A second production route is provided in

```
CGJteamLab/Wyler/Proposition11_26.lean
```

by

```
Geometry.euclid_proposition_11_26_wyler_core  
Geometry.euclid_proposition_11_26_wyler
```

The Hilbert–Wyler route does not duplicate the long metric and order proof. Instead it reroutes the common spatial incidence fields through `HilbertWylerAxioms` and reuses the same XI.26 reconstruction.

No coordinates, vectors, scalar products, numerical angle measures, or trigonometric functions occur in either production route.

2 Euclid’s Statement

Euclid’s Proposition XI.26 states:

To construct a solid angle equal to a given solid angle on a given straight line, and at a given point on it.

Let AB be the prescribed straight line and A the prescribed point on it. Let the source solid angle have vertex D and be bounded by the three plane angles

$$\angle EDC, \quad \angle EDF, \quad \angle FDC.$$

The required construction is a solid angle at A , one of whose edges lies on the prescribed ray from A through B , and whose three face angles are congruent to the three source face angles.

The formal theorem uses the project predicate

```
HilbertTrihedralConfiguration Geo D E C F
```

for the source trihedron and concludes the existence of points L, H such that

```
HilbertTrihedralRealizesThreeAngles  
Geo A B L H E D C C D F F D E
```

holds. Thus the three target face angles are

$$\angle BAL, \quad \angle LAH, \quad \angle HAB,$$

and they correspond respectively to

$$\angle EDC, \quad \angle CDF, \quad \angle FDE.$$

3 Classical Configuration

The source configuration has vertex D and three rays DE, DC, DF . Euclid chooses a point F on the third ray and drops the perpendicular FG to the plane determined by ED and DC . The foot is G , so

$$G \in \text{plane}(E, D, C), \quad FG \perp \text{plane}(E, D, C).$$

The segment DG is then joined.

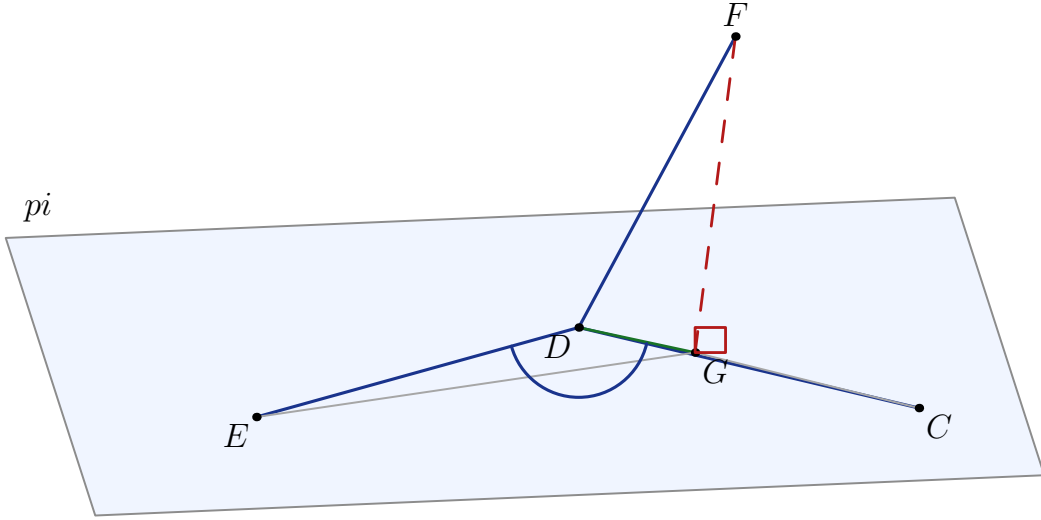


Figure 1: The source trihedral angle at D . The point F on the third spatial edge is projected perpendicularly to the base plane π through E, D, C , with foot G . The formal proof does not assume any particular order position of G inside that plane.

At the prescribed point A on the line AB , Euclid constructs a plane angle equal to $\angle EDC$ and a second plane angle equal to $\angle EDG$. On the second constructed ray he chooses K with

$$AK = DG.$$

Through K he erects a line perpendicular to the target base plane and chooses H on that line so that

$$KH = GF.$$

The metric verification is then driven by right triangles. Euclid cuts off on the first target ray a segment equal to DE , joins the necessary segments, and repeatedly applies planar triangle congruence. The right triangles based on DG, GE, GC are matched with the corresponding target right triangles based on AK and the copied planar distances. This ultimately gives

$$\angle BAH = \angle EDF, \quad \angle HAL = \angle FDC,$$

while the copied base construction already gives

$$\angle BAL = \angle EDC.$$

The classical diagram depicts one generic position of G . The formal theorem cannot use that diagrammatic convention as a hidden hypothesis.

4 Euclid's Classical Proof

Euclid's proof can be organized into five steps.

4.1 Project the third edge to the base plane

A point F is chosen on the third source ray and FG is drawn perpendicular to the plane through E, D, C , using Proposition XI.11. The point G is the foot of this perpendicular.

The defining property of a line perpendicular to a plane implies that FG is perpendicular to every relevant line in the base plane through G .

4.2 Copy the base angle and the projected direction

At A , on the prescribed line AB , Euclid constructs

$$\angle BAL = \angle EDC$$

and

$$\angle BAK = \angle EDG.$$

He then lays off

$$AK = DG.$$

In the classical text this is presented as a direct application of planar angle and segment construction. The production Lean proof packages the same information differently: it first copies a nondegenerate base triangle and then constructs the point corresponding to G by its position relative to the copied base angle.

4.3 Erect the target normal

At K Euclid erects a line perpendicular to the plane through B, A, L , using XI.12, and chooses H on this normal such that

$$KH = GF.$$

Thus the source and target configurations have equal perpendicular heights above their respective base planes.

4.4 Recover the spatial distances

Because FG and KH are perpendicular to their respective base planes, the relevant triangles are right triangles. Euclid first establishes the copied base distances and then applies I.4 repeatedly to obtain equal spatial distances from H and F to the corresponding base vertices.

Schematically,

$$DG = AK, \quad GF = KH$$

and equality of the appropriate base distances give the equal hypotenuses needed for the final face comparisons.

4.5 Recover the remaining face angles

The final use of I.8 identifies the remaining plane angles by SSS. Hence all three face angles of the constructed trihedron at A are equal to the corresponding face angles of the source trihedron at D .

The classical proof is therefore fundamentally synthetic: orthogonal projection, planar copying, perpendicular erection, SAS/SSS triangle congruence, and no numerical representation of solid angles.

5 Why XI.26 Is Structurally Difficult

The main formal difficulty is not the final metric verification. Once a correct target copy of the projected base point G has been constructed, the perpendicular lift and SSS recovery are systematic.

The difficult point is that Euclid's diagram suppresses the complete order type of G in the base plane.

The source projection theorem produces

$$G \in \pi, \quad FG \perp \pi, \quad D, E, C \in \pi,$$

where π is the base plane. It does *not* say that DG lies inside $\angle EDC$. A complete synthetic proof must therefore distinguish all possible ray positions.

The production theorem handles:

- $G = D$;
- G on the ray DE ;
- $E - D - G$, the ray opposite DE ;
- G on the ray DC ;
- $C - D - G$, the ray opposite DC ;
- the first adjacent exterior sector;
- the second adjacent exterior sector;
- the interior sector;

- the cyclic/opposite sector.

The noncollinear part of this classification is expressed by the reusable interface theorem

```
hilbert_space_three_rays_order_or_cyclic_in_plane
```

while collinear boundary positions are handled by

```
hilbert_space_collinear_ray_position_in_plane
```

This is the central formal refinement of Euclid XI.26: the classical picture shows a representative configuration; the Lean theorem proves the construction independently of where the perpendicular foot happens to lie.

6 Formal Reconstruction: Projection, Base Copy, and Lift

6.1 The copied base triangle

The first production structure is

```
HilbertXI26BaseCopy
```

For a source base triangle D, E, C and prescribed target ray AB , it records a target plane σ and points B_0, L such that:

- $A, B, B_0, L \in \sigma$;
- B_0 lies on the same ray from A as B ;
- $AB_0 \cong DE$;
- $AL \cong DC$;
- $B_0L \cong EC$;
- the copied base triangle is nondegenerate;
- $\angle B_0AL \cong \angle EDC$;
- the same angle equality is available with the prescribed ray AB .

The local theorem

```
hilbert_XI26_copy_base_triangle_in_plane
```

performs the construction in an already chosen target plane. The wrapper

```
hilbert_XI26_copy_base_triangle_on_ray
```

chooses such a plane through the prescribed ray.

The use of the auxiliary point B_0 is deliberate. The theorem must keep the original prescribed point B as the direction of the first target ray, but there is no reason for the original segment AB itself to have length DE . The construction therefore lays off the correct length on the same ray.

sigma

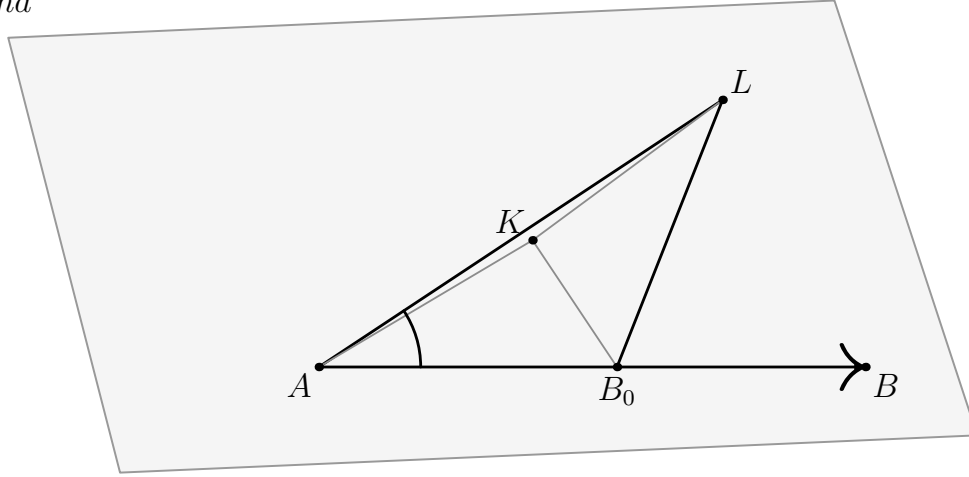


Figure 2: The copied planar base at the prescribed point A . The point B_0 lies on the prescribed ray AB , L completes the copied base triangle, and K represents the copied projection point. Metric equalities and angle congruences are stated in the text rather than written on the diagram.

6.2 The source projection

The theorem

```
hilbert_XI26_source_projection
```

constructs the base plane π , a normal line, and the foot G . Its output includes

$$D, E, C, G \in \pi, \quad F \notin \pi, \quad F \in \text{normal}, \quad F \neq G,$$

together with

$$\text{HilbertLinePerpendicularPlaneAt}(\text{normal}, \pi, G).$$

This stage uses the already established Proposition XI.11.

6.3 The interior model

When the ray DG meets the open segment EC , the target point can be constructed inside the copied base angle. The structure

```
HilbertXI26InteriorPointCopy
```

records a target point K with

$$DG \cong AK, \quad EG \cong B_0K, \quad CG \cong LK,$$

together with the corresponding two component angle congruences.

The combined theorem

```
hilbert_XI26_copy_base_with_interior_point
```

packages the copied base and the copied interior projection point.

The full interior branch is

```
euclid_proposition_11_26_of_interior_projection
```

6.4 Boundary and exterior positions

The remaining projection positions cannot be forced into the interior construction without losing orientation information. The production proof therefore contains separate theorems for the boundary rays, opposite rays, two adjacent exterior sectors, and the cyclic sector.

The terminal branch theorems are:

```
euclid_proposition_11_26_of_projection_foot_eq_vertex  
euclid_proposition_11_26_of_projection_on_first_ray  
euclid_proposition_11_26_of_projection_on_second_ray  
euclid_proposition_11_26_of_first_exterior_projection  
euclid_proposition_11_26_of_second_exterior_projection  
euclid_proposition_11_26_of_cyclic_projection  
euclid_proposition_11_26_of_projection_opposite_first_ray  
euclid_proposition_11_26_of_projection_opposite_second_ray
```

The exterior cases are where the reusable spatial Hilbert-T15 infrastructure is needed. The proposition-specific file builds the local source and target extension data; the proposition-independent angle-addition and same-side machinery now lives in

```
CGJteamLab/HilbertInterfaceXI.lean
```

6.5 The spatial lift

Once a target base point K has been constructed with the three required base distances, the remaining step is common in spirit to all branches. XI.12 erects a target normal at K , the source height GF is copied to that normal, and right-triangle congruence recovers the spatial distances.

The two principal lifting interfaces in the proposition file are

```
hilbert_XI26_lift_base_point_copy  
hilbert_XI26_lift_base_point_copy_ne
```

They culminate in a point H for which

```
HilbertTrihedralRealizesThreeAngles
Geo A B L H E D C C D F F D E
```

holds.

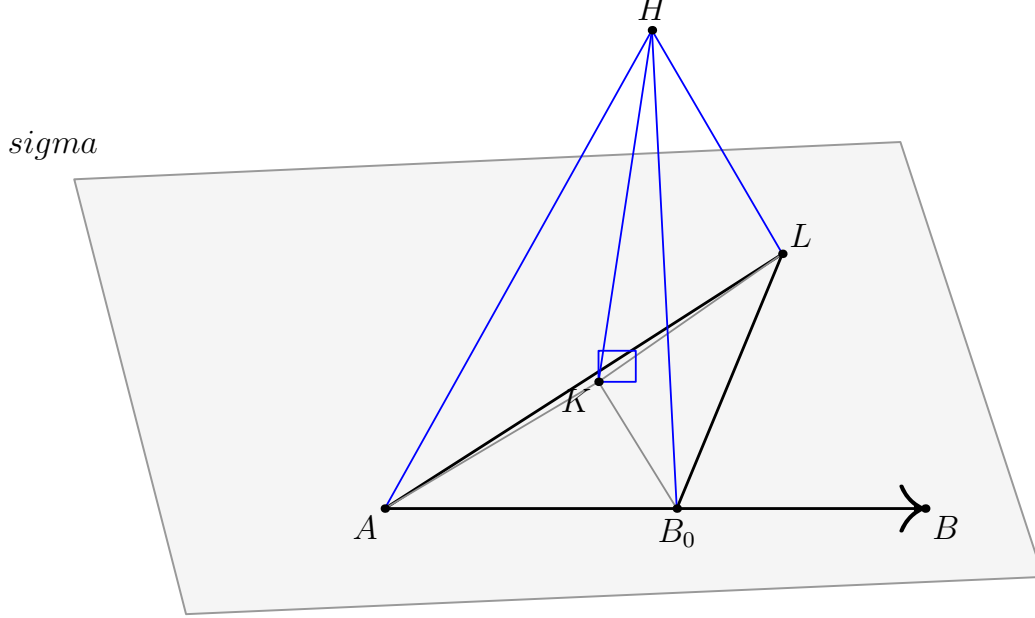


Figure 3: The final spatial lift. The points A, B_0, L, K lie in the target base plane, while H is obtained on the perpendicular through K . The three face-angle congruences with the source trihedron are recovered by the shared synthetic metric argument.

6.6 The complete dispatcher

The public direct theorem is:

```
theorem euclid_proposition_11_26
  [Hinc : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [HSO : HilbertSpaceOrder
    (Geo := Geo) (H := Hinc) (S := S)]
  [HSC : HilbertSpaceCongruence
    (Geo := Geo) (H := Hinc) (S := S)]
  [HSE : HilbertSpaceEuclidean Geo]
  (D E C F A B : Geo.Point)
  (hTri : HilbertTrihedralConfiguration Geo D E C F)
  (hAB : Ne A B) :
  exists L H : Geo.Point,
    HilbertTrihedralRealizesThreeAngles
      Geo A B L H E D C C D F F D E
```

The final proof first constructs the source projection and then performs a complete order dispatch. The dispatcher introduces no new geometric construction; every branch has already been solved

by a dedicated theorem.

7 The Hilbert–Wyler Route for XI.26

The separate production file

```
CGJteamLab/Wyler/Proposition11_26.lean
```

implements XI.26 through a Hilbert–Wyler incidence route.

The architectural situation is notably cleaner than in XI.24 and XI.25. There is no second copy of the approximately five-thousand-line XI.26 development. The audited direct proof is already parametric in

```
HilbertSpaceIncidence Geo
```

and proposition-independent geometry has already been moved to `HilbertInterfaceXI`. The Wyler route can therefore change the incidence package supplied to the existing theorem while leaving the whole order, congruence, `PlaneGeo`, right-angle, SAS, SSS, and T15 core unchanged.

7.1 The routed incidence package

The theorem

```
hilbertSpaceIncidence_routed_through_wyler
```

constructs a new `HilbertSpaceIncidence` value from:

- the existing Hilbert 3D spatial incidence package;
- an explicit `HilbertWylerAxioms Geo` package.

The common incidence fields are taken from Wyler:

```
two_points_on_each_line  
plane_through  
plane_unique  
line_in_plane
```

The compatibility facade also requires a point on each plane. The Wyler package provides the stronger fact that every plane contains three noncollinear points, so

```
point_on_each_plane
```

is derived from

```
W.three_noncollinear_on_plane
```

Two genuinely three-dimensional Hilbert fields are retained from the ambient 3D package:

```
plane_second_common_point
four_noncoplanar
```

Thus the routing has the conceptual form

$$\begin{array}{ccc} \text{Hilbert-Wyler common incidence} & + & \text{Hilbert 3D residual incidence} \\ \text{I.3-I.6 level data} & & \text{I.7-I.8 level data} \end{array} \implies \text{XI.26 incidence package.}$$

This is an incidence refactor, not a claim that the present XI.26 theorem is already dimension-free.

7.2 The explicit Wyler core

The theorem

```
euclid_proposition_11_26_wyler_core
```

assumes an explicit

```
[W : HilbertWylerAxioms Geo]
```

It constructs the routed incidence value

```
HSIW : HilbertSpaceIncidence Geo
```

and invokes the already proved direct theorem with

```
(HSI := HSIW)
```

This point is mathematically significant. The Wyler theorem is not an alias of the final direct conclusion: the direct XI.26 proof is re-instantiated with a different spatial incidence record. Consequently the common incidence operations used by the proof are definitionally supplied through Wyler.

At the same time the theorem deliberately retains the established spatial order and congruence classes. It does not yet construct a local metric 3-flat in an arbitrary-dimensional ambient geometry.

7.3 The public Hilbert 3D corollary

The public theorem

```
euclid_proposition_11_26_wyler
```

does not add `HilbertWylerAxioms` as an extra public geometric assumption. It first defines

```
let W : HilbertWylerAxioms Geo :=  
  hilbertWylerAxioms_of_hilbert3D  
    (Geo := Geo)
```

and then applies the explicit Wyler core.

Thus, in the present three-dimensional library,

$$\text{Hilbert 3D} \implies \text{Hilbert--Wyler incidence} \implies \text{XI.26.}$$

The result is parallel to the architecture already used for XI.23, but XI.26 requires dramatically less duplicated proposition code because its direct proof has already been refactored around reusable interfaces.

8 Proof Architecture and Formal Obligations

The complete reconstruction is most naturally read as one continuous construction.

1. The source trihedron determines the nondegenerate base angle EDC and guarantees that F lies outside the base plane.
2. XI.11 constructs the perpendicular projection $F \rightarrow G$ onto the source base plane.
3. The base triangle DEC is copied to a target plane through the prescribed ray AB , producing B_0, L .
4. The position of G relative to DE and DC is classified completely.
5. In the corresponding target sector a point K is constructed so that the three distances from G to D, E, C are copied by the three distances from K to A, B_0, L .
6. XI.12 erects a normal through K to the target base plane.
7. The height GF is copied to the target normal.
8. Spatial right-triangle congruence gives the three lateral distances.
9. Spatial SSS recovers the two remaining face-angle congruences.
10. The target noncoplanarity and the three angle congruences are packaged as `HilbertTrihedralRealizesThr`

Several obligations that are invisible in Euclid's diagram are explicit in Lean:

- every use of planar order occurs in an explicit `PlaneGeo`;
- every target point must be proved to remain in the chosen target plane;

- ray orientation must be preserved under each segment construction;
- opposite-ray and exterior-sector cases require explicit same-side and opposite-side information;
- angle addition requires a synthetic T15 interface rather than numerical angle measures;
- the final four points of the target trihedron must be proved noncoplanar;
- the prescribed point B fixes a ray, not the target length, hence the auxiliary point B_0 .

These obligations explain the formal length of XI.26. They are not numerical overhead; they are the explicit incidence and order content hidden by the classical drawing.

9 Dependencies and Reuse

The production direct file imports

```
CGJteamLab.HilbertInterfaceXI
CGJteamLab.Proposition11_11
CGJteamLab.Proposition11_12
```

XI.11 supplies the source perpendicular from F to the base plane. XI.12 supplies the target perpendicular through the copied projection point K .

The remaining substantial machinery is reusable interface mathematics rather than proposition-local dependence on earlier Book XI theorem files. In particular, the post-audit `HilbertInterfaceXI` contains the general results for:

- points off lines inside planes;
- spatial right-angle recovery from a plane normal;
- lifted-distance comparison;
- spatial SSS angle recovery;
- ray-position classification;
- opposite-extension and same-side transport;
- spatial segment-construction uniqueness;
- the spatial Hilbert-T15 angle-addition chain.

A notable consequence of the audit is that XI.26 no longer imports `Proposition11_23`. The planar three-ray classification originally developed during XI.23 has been promoted to the common interface.

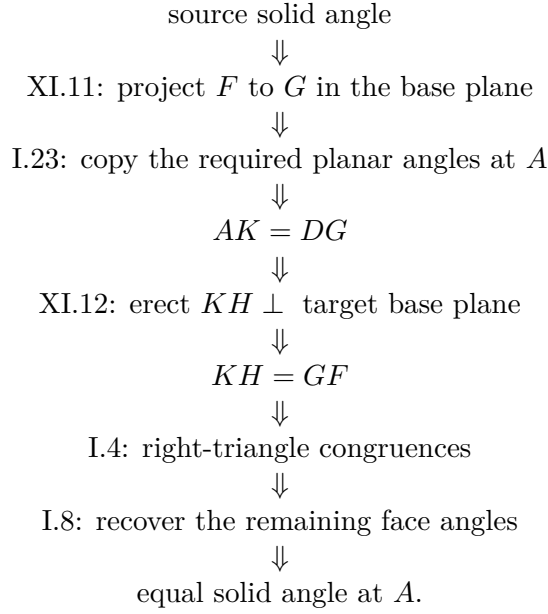
The Wyler production file imports

```
CGJteamLab.Proposition11_26
CGJteamLab.HilbertWyler3DCompatibility
```

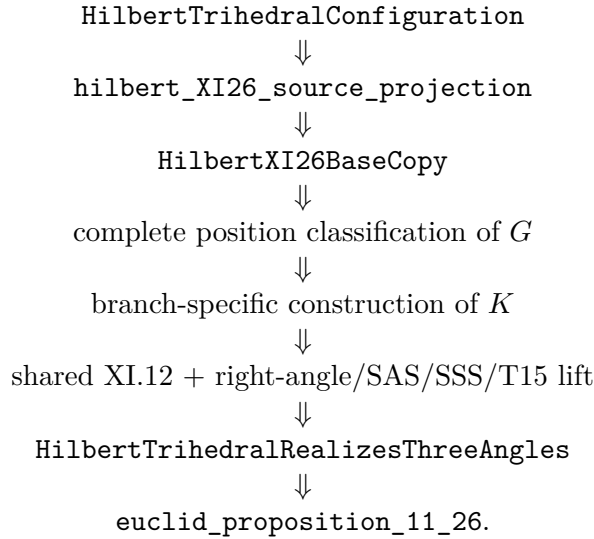
and reuses the complete direct metric/order proof after changing the supplied incidence record.

10 Classical and Formal Dependency DAGs

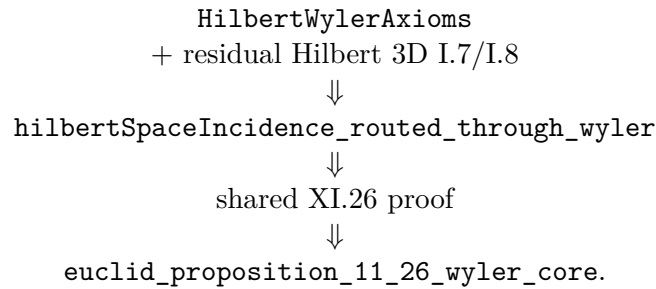
The classical construction has the compact chain



The formal direct chain is longer:



The Hilbert–Wyler production route inserts an incidence factorization before the same direct theorem:



For the public 3D theorem,

$$\text{Hilbert 3D} \implies \text{hilbertWylerAxioms_of_hilbert3D} \implies \text{euclid_proposition_11_26_wyler.}$$

11 Comparison with Euclid

At the highest level the formal reconstruction follows Euclid closely:

1. choose a base plane of the source solid angle;
2. project the third edge orthogonally to that plane;
3. reproduce the projected planar geometry at the prescribed ray;
4. erect a perpendicular at the copied projection point;
5. copy the perpendicular height;
6. recover the remaining face angles from congruent triangles.

The main difference is not metric but order-theoretic. Euclid draws one position of G and proceeds as if the required planar construction has a single visible orientation. The formal theorem must be valid for every trihedral configuration satisfying its hypotheses. Consequently it proves a complete classification of G and supplies a construction in every sector.

A second difference is representational. Euclid directly says that the required plane angles are constructed at A . The formal proof often represents that construction by a copied triangle or by three copied distances. SSS then recovers the angle statement. This is not a change in geometry; it is a more stable synthetic interface for a proof assistant.

A third difference is architectural. The completed audit separates proposition-specific construction from reusable spatial mathematics. General order and T15 lemmas no longer live inside the XI.26 file merely because they were discovered while proving XI.26.

The Hilbert–Wyler route adds a further structural observation: much of the spatial incidence needed by XI.26 belongs to the common Wyler-compatible incidence layer, while the present theorem still retains specifically three-dimensional residual incidence and metric structure.

12 Diagrammatic Audit

Three figures are mathematically justified for the final documentation.

1. **Source projection.** The source trihedron at D , its base plane through E, D, C , the point F , and the perpendicular projection $F \rightarrow G$. This figure should make clear that the order position of G is not fixed by the projection theorem.
2. **Copied planar configuration.** The prescribed ray AB , the auxiliary point B_0 on the same ray, the copied base point L , and a representative copied projection point K . The figure should distinguish the prescribed direction AB from the laid-off segment AB_0 .
3. **Final spatial lift.** The target base plane through A, B_0, L, K , the perpendicular through K , the lifted point H , and the three target face angles corresponding to the source trihedron.

No separate figure is required for each of the nine order branches. Those branches represent different logical positions of the same planar datum, not nine geometrically stable constructions. A small schematic ray-order diagram may be useful inside the second figure, but the documentation should avoid a page of near-duplicate branch pictures.

13 Audit Status

This documentation is based on the current production sources

```
CGJteamLab/Proposition11_26.lean
CGJteamLab/Wyler/Proposition11_26.lean
```

Both production modules have been compiled successfully after the XI.26 architecture audit.

The direct proposition file was reduced by moving proposition-independent mathematics to `HilbertInterfaceXI`. In particular, the spatial Hilbert-T15 chain and the three-ray order classification are now shared interface results. Redundant and superseded local helpers were removed, and XI.26 no longer imports Proposition XI.23 merely to obtain a general planar order theorem.

The Hilbert–Wyler production theorem was then checked by constructing a routed `HilbertSpaceIncidence` package and running the already compiled XI.26 theorem against it. This confirms that the common incidence portion of the current proof can be supplied through `HilbertWylerAxioms` without duplicating the metric/order development.

A separate kernel `#print axioms` comparison for

```
Geometry.euclid_proposition_11_26
Geometry.euclid_proposition_11_26_wyler_core
Geometry.euclid_proposition_11_26_wyler
```

is not recorded here until that audit is run explicitly. The source architecture already shows that the public Wyler theorem adds no new *typeclass assumption*: its Wyler package is derived from the established Hilbert 3D environment.

14 Conclusion

Proposition XI.26 is a constructive theorem about transporting a complete trihedral angle to a prescribed spatial location and direction. Euclid’s classical argument is short because the diagram fixes an apparently generic projection geometry. Formal reconstruction shows that the decisive hidden content is the order type of the projection foot.

The production proof therefore separates the problem into three layers:

source projection $\implies$ complete planar position analysis $\implies$ common spatial lift and face recovery.

This organization preserves Euclid’s synthetic mechanism while making every incidence, order, orientation, and nondegeneracy obligation explicit.

The Hilbert–Wyler route gives a second structural conclusion. XI.26 does not require a second proposition-scale proof in order to expose its Wyler-compatible incidence content. After the direct development is properly modularized, the same theorem can be run with a spatial incidence record whose common fields are supplied by `HilbertWylerAxioms`, while the residual 3D Hilbert incidence and metric layers remain unchanged.

Thus XI.26 provides a particularly clean example of the role of the Wyler program in the project:

change the incidence route where it genuinely changes + reuse the established synthetic metric core.

It also prepares the next proposition structurally. XI.27 begins by invoking XI.26 to reproduce the solid angle of a given parallelepiped on a prescribed straight line; the completed formal XI.26 theorem now supplies that construction without any hidden assumption on the position of the projected foot.